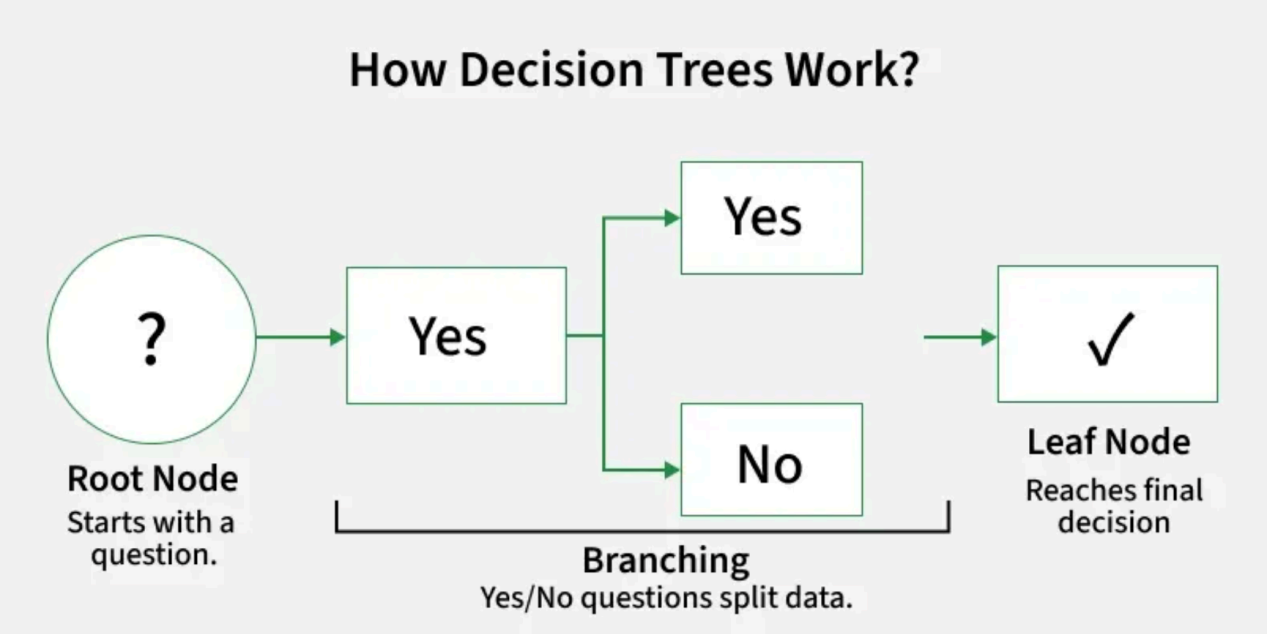
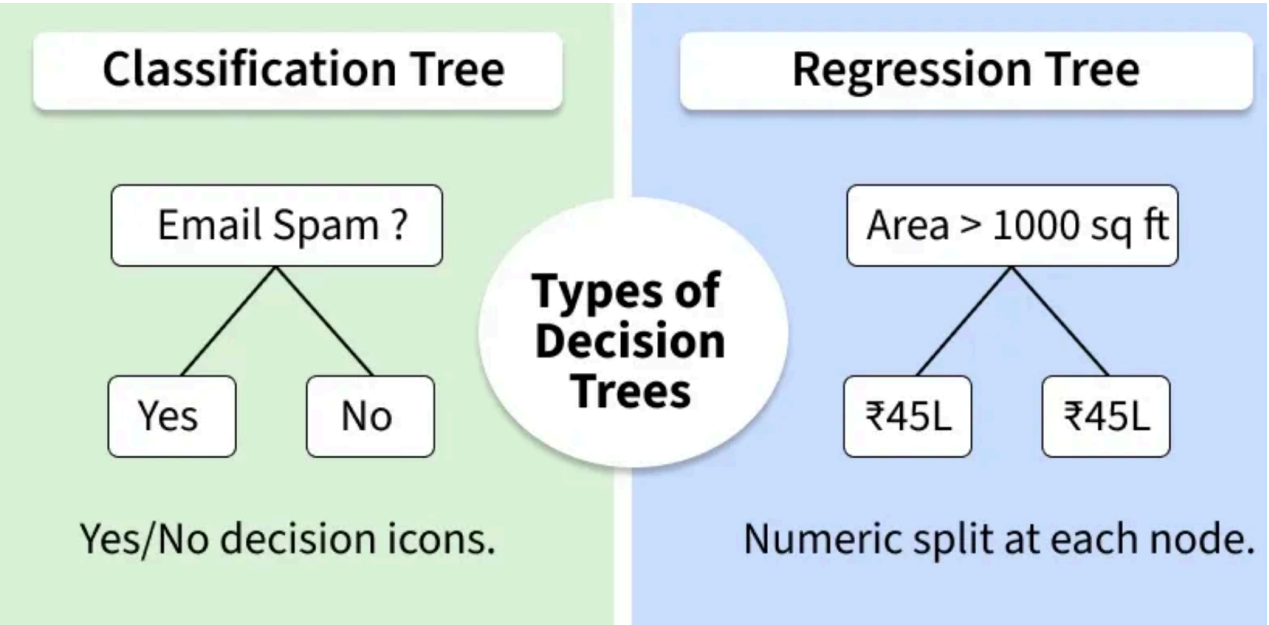
Decision Tree

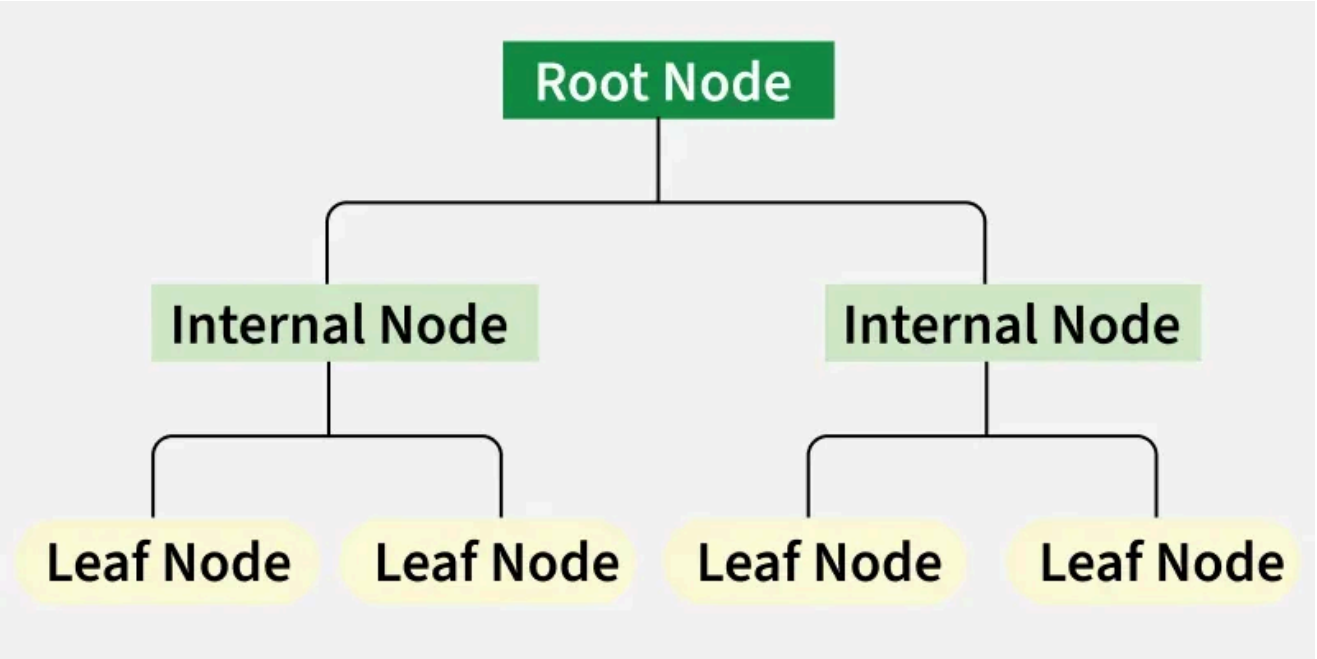
- Can perform both **Classification** and **Regression** task
- A Decision Tree helps us to make decisions by **mapping out different choices and their possible outcomes** (G4G).





- it is the fundamental component of **Randomforest**
- Require little data preparation = do not require feature scalling or centering at all
- **CART = Classification and Regression Tree, it is a greedy algorithm**
  - A **greedy algorithm** often produces such a reasonably good predictions or results
- The prediction are made by the DecisionTree from the root to the leaf
- to avoid the **overfitting during the training**, you need to restrict the DecisionTree's Freedom

Overview of DecisionTree Structure



- Root node = Starting point representing the whole dataset
- Branches = Lines connecting nodes showing the flow from one decision to another
- Internal Nodes = Points where decisions are made based on data features.
- Leaf Nodes = End points of the tree where the final decision or prediction is made

How it works

1. **Start with the Root Node:** It begins with a main question at the root node which is derived from the dataset's features.
2. **Ask Yes/No Questions:** From the root, the tree asks a series of yes/no questions to split the data into subsets based on specific attributes.
3. **Branching Based on Answers:** Each question leads to different branches:
   - If the answer is yes, the tree follows one path.
   - If the answer is no, the tree follows another path.
4. **Continue Splitting:** This branching continues through further decisions helps in reducing the data down step-by-step.
5. **Reach the Leaf Node:** The process ends when there are no more useful questions to ask leading to the leaf node where the final decision or prediction is made.

Splitting Criteria

the splitting criteria used by the regression tree and the classification tree are different.
for classification tree: the goal is to divide the data into smaller, more **homogenous** (most of the samples at **each node** are from one class) groups.
The process of Splitting data at each node is important.
it finds the best feature splits the data on

- **Gini Impurity**: This criterion measures how "impure" a node is. The **lower** the **Gini Impurity** the **better** the feature **splits the data into distinct categories**.

$$Gini\ Index = \Sigma_i p_i(1 - p_i)$$

where p_i is the probability of class i and the interval of Gini is $[0, 0.5]$. For a two-class problem, the Gini impurity for a given node is:

$$p_1(1 - p_1) + p_2(1 - p_2)$$

- **Entropy:** This measures the amount of **uncertainty** or **disorder** in the data. The tree tries to reduce the **entropy** by **splitting the data on features that provide the most information about the target variable**.

Pruning = cutting

- It is a crucial technique used to prevent **overfitting** in decision Tree.
- This technique **reduces the complexity of the tree** by **removing branches that have little predictive power**. It improves model performance by **helping the tree generalize better to new data**. It also makes the model simpler and faster to deploy.
- It is useful when a Decision Tree is too deep and starts to capture noise in the data.

$$\sum_{m=1}^{|T|} \sum_{i:\ x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

Note: **Pruning the tree back** could reduce the variance but that would add more bias. So dont prune it back, but get more bushy trees tend to have low bias and get rid of the variance by averaging them instead.

Pros and Cons of Trees

Pros

- Easy to interpret, easier to explain than the linear Regression
- Some peoplpe believe that decision trees more closely mirror human decision-making than do the regression and classification approaches
- Trees can be display graphically, and are easily interpreted even by a non-expert

- Tree can easily handle qualitative predictors without the need to create dummy variables **Cons**
- trees generally dont have the same level of predictive accuracy as some of the other regression and classification approaches - It can be computationally expensive especially with a large number of trees.
- It's harder to interpret the model compared to simpler models like decision trees.- tree can be very non-robust, a small change in the data can cause large change in the final estimated tree.
- It is more likely to overfit when train on small dataset.